

MARBLE: Lightweight EEG-to-fMRI Translation via Mamba-Attention and ROI-Conditioned Decoding

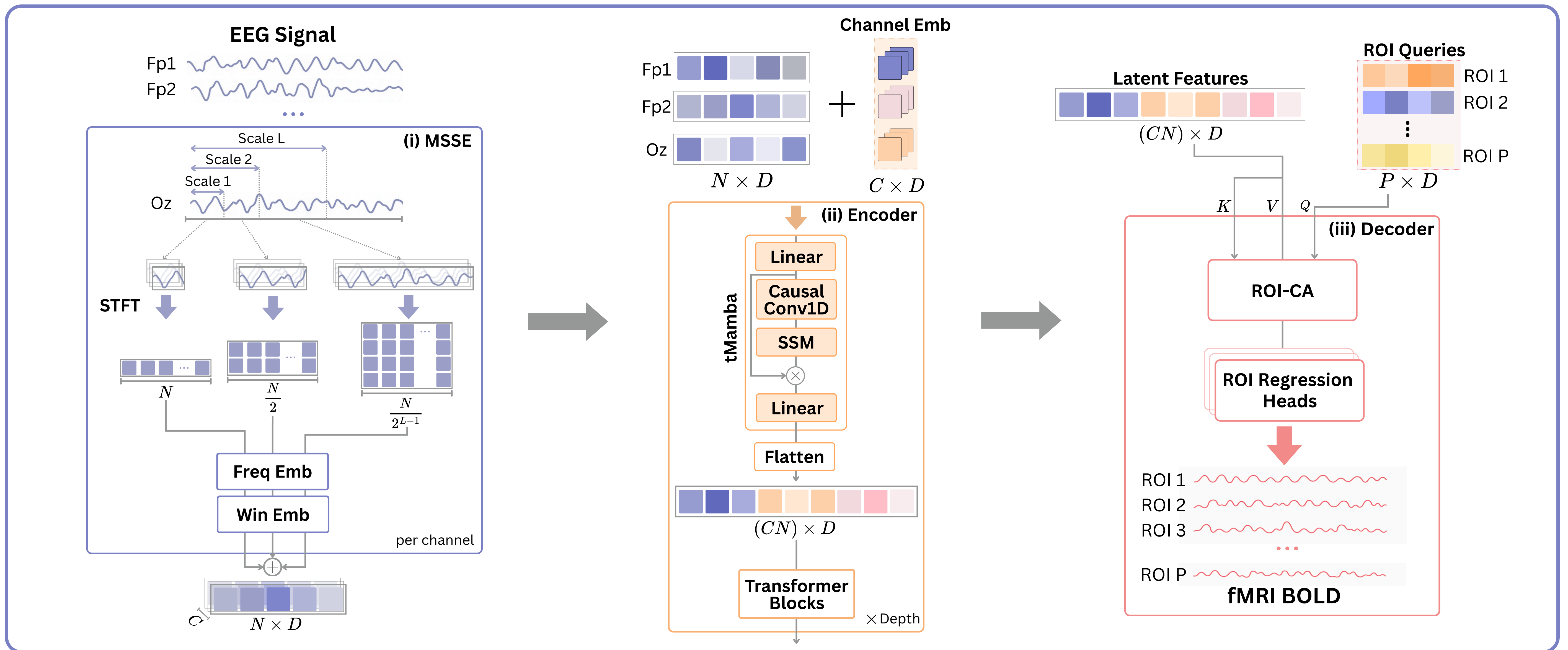
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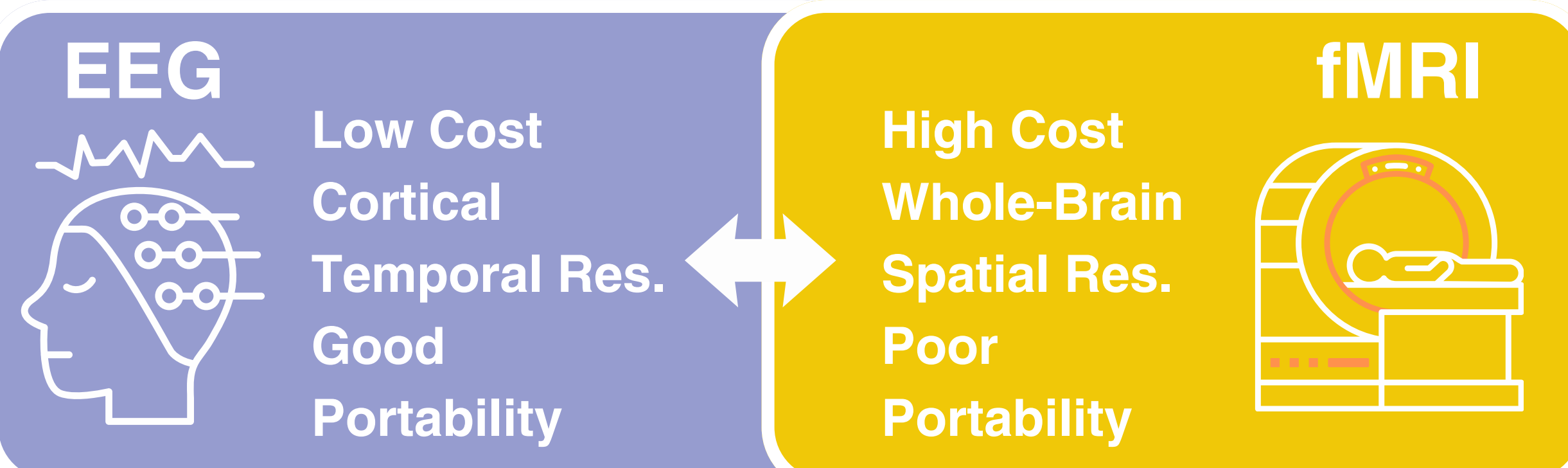
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Figure 1. Overview of the Proposed Architecture



I. Introduction: Lightweight EEG-to-fMRI Translation



- **fMRI** and **EEG** provide complementary spatial and temporal information
- **EEG** captures rapid neural dynamics and offers a promising basis for estimating fMRI BOLD signals through cross-modal translation.

II. Contributions

- ▶ Efficient **Mamba-Transformer** encoder for HRF-related temporal modeling, reducing parameters by 3.2x and memory by 2.3x.
- ▶ Interpretable **ROI-wise cross-attention** decoders for region-specific BOLD prediction.
- ▶ Improved efficiency–performance trade-off while preserving functional connectivity patterns.

III. Materials and Experimental Settings

Simultaneous EEG-fMRI Dataset [1] → subjects rested with eyes closed

- 29 scans from 22 subjects, each scans has a duration of ~20 mins
- **Evaluation Strategy:** All scans were randomly divided into train/val/test in 3:1:1 ratio. No subject overlap between sets
- fMRI time-series were extracted using DiFuMo [2] atlas
- For each fMRI time point, a 16-second of preceding EEG was extracted [1]

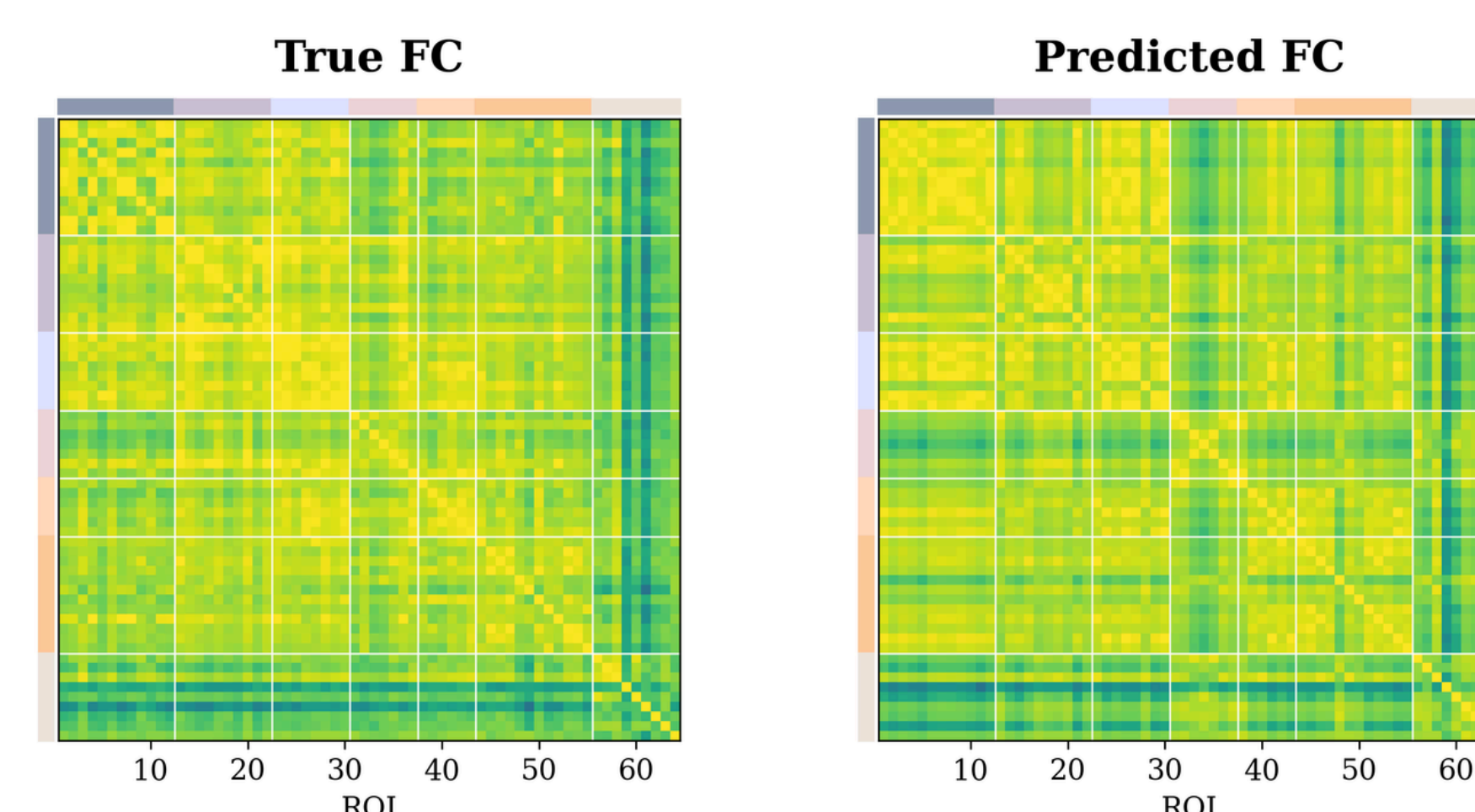
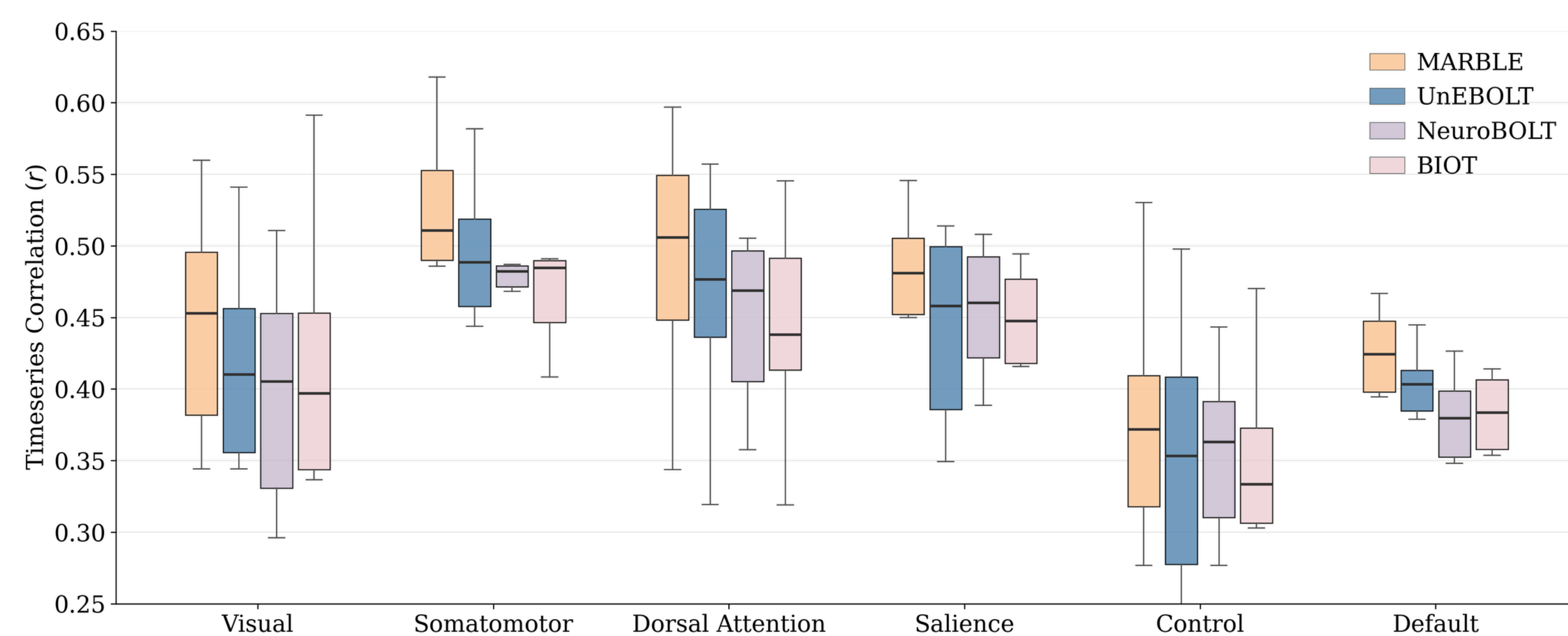
Metrics: Signal-Level: Correlation Coefficient (r), MSE; Connectivity-Level: FC-Corr, FC-F1*

* FC matrices = Pearson correlations across ROI time series, F1 is computed by thresholding at the 75th percentile

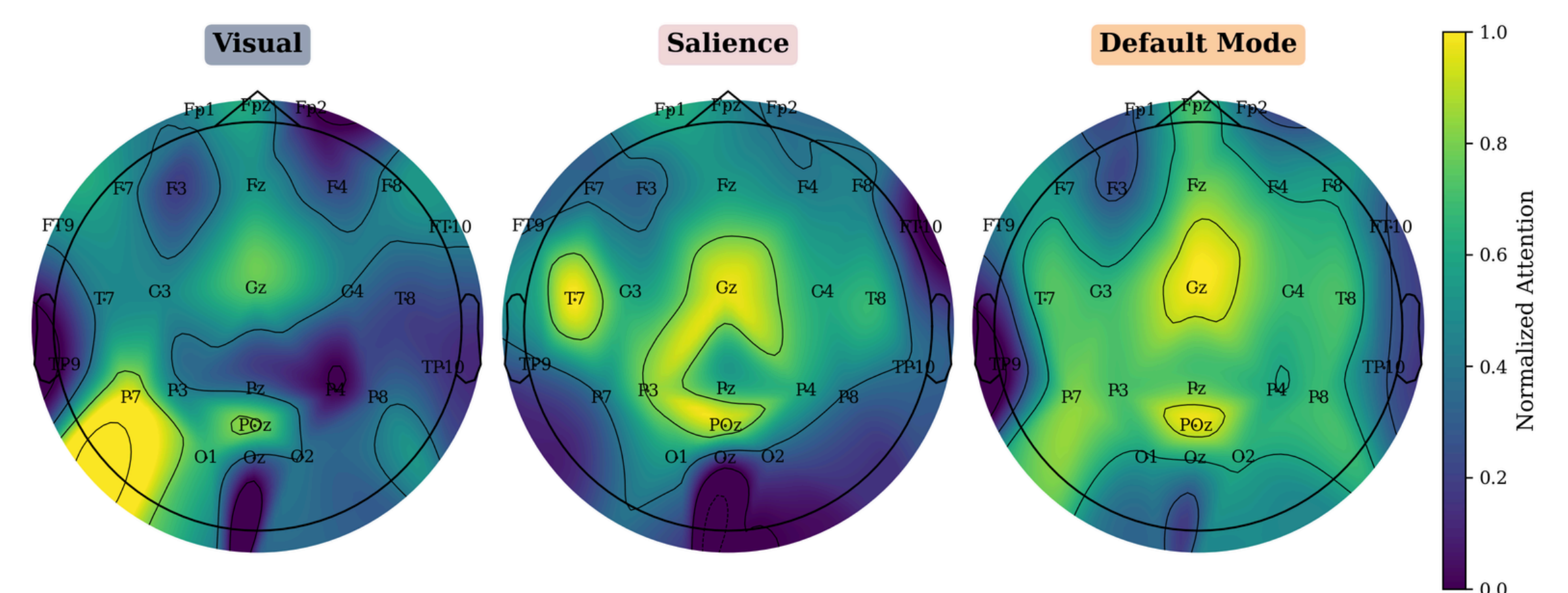
Table 1. Quantitative comparison on 6 unseen test scans from the VU dataset. Results are reported as mean $\pm$ standard deviation. Bold indicates best performance. Gray-shaded rows indicate ablation studies.

Model	Corr (r) $\uparrow$	MSE $\downarrow$	FC-Corr $\uparrow$	FC-F1 $\uparrow$	Params $\downarrow$	Mem $\downarrow$
BIOT	0.396 $\pm$ 0.041	0.229 $\pm$ 0.020	0.439 $\pm$ 0.144	0.404 $\pm$ 0.025	2.0M	4.4 GB
LaBraM	0.179 $\pm$ 0.075	0.250 $\pm$ 0.009	0.359 $\pm$ 0.114	0.429 $\pm$ 0.006	5.8M	11.5 GB
NeuroBOLT	0.406 $\pm$ 0.038	0.223 $\pm$ 0.017	0.469 $\pm$ 0.115	0.418 $\pm$ 0.012	7.9M	13.7 GB
UnEBOLT	0.412 $\pm$ 0.049	0.218 $\pm$ 0.019	0.469 $\pm$ 0.093	0.427 $\pm$ 0.016	8.2M	13.8 GB
MARBLE	0.434 $\pm$ 0.047	0.210 $\pm$ 0.020	0.456 $\pm$ 0.129	0.454 $\pm$ 0.024	2.6M	6.0 GB
W/O ROI CA	0.413 $\pm$ 0.040	0.218 $\pm$ 0.019	0.395 $\pm$ 0.084	0.431 $\pm$ 0.018	-	-
W/O tMamba	0.424 $\pm$ 0.042	0.211 $\pm$ 0.018	0.440 $\pm$ 0.125	0.442 $\pm$ 0.019	-	-

IV. Results



- ▶ Consistent performance across diverse brain systems
- ▶ Superior signal-level reconstruction
- ▶ Substantially fewer parameters and lower memory usage
- ▶ Functional organization preservation



- Visual showed posterior contributions, consistent with visual-related activity in resting EEG [3], Saliency and DMN shared midline–posterior components

V. Conclusion

Mamba-Transformer encoder provides a favorable balance between reconstruction performance and computational efficiency. ROI-conditioned cross-attention reveal network-specific electrode contributions aligned with known EEG–BOLD coupling. Study is limited by the use of a single site cohort of 22 subjects. Future work will extend MARBLE to larger multi-site cohorts and task-based fMRI.

[1] Li, Y. et al., NeuroBOLT, NeurIPS (2024); [2] Dadi, K. et al., Fine-grain atlases for fMRI analysis, NeuroImage (2020); [3] Goldman, R.I. et al., Simultaneous EEG and fMRI, Neuroreport (2002)